

“Call Me Before I Open the Fridge”
Designing Conversational Support for Binge Eating Risk in Solitary Eating

Ryuhaerang Choi • Seohyeon Yoo • Sung-Ju Lee

UbiComp ISWC 2026

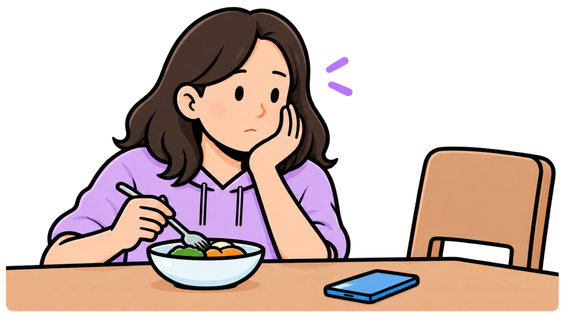
KAIST

TL;DR

To inform conversational support design for binge eating risk during solitary eating, we conducted design sessions with 17 individuals with eating disorders. Participants expressed differing expectations for proactive intervention and user control. Some envisioned eating-focused conversation to support planning and reflection, while others preferred redirection to avoid intensifying unwanted thoughts.

01 Background

55.5 million people estimated to be living with eating disorders worldwide.



binge eating
[ˈbɪndʒ ˌiːtɪŋ]
Eating unusually large amounts in a short time while feeling unable to stop or control eating.

Why solitary eating matters

- **Common binge eating contexts.** Binge eating episodes often occur when people with eating disorders eat alone.
- **Embarrassment about eating.** Some people eat alone because they feel embarrassed about others seeing how much they eat.
- **Everyday support gaps.** Timely help is often unavailable when eating difficulties arise.

Why conversational support

Conversation could adapt support as eating unfolds. When people eat alone, an agent may offer a sense of company without competing with an existing conversation.

When it adds value beyond a brief prompt and how it should begin, adapt, or end remain unclear.

02 Research Goal

> How do people with eating disorders **envision conversational support** during solitary eating?

03 Method

Participants 17 adults; 13 bulimia nervosa · 4 binge eating disorder

Individual design sessions online · ~30 min

- **Experience sharing** Experiences and challenges of eating alone
- **Support needs** When and which support might help as urges changed
- **Envisioned AI support** Timing, topics, interaction mode, and user control

04 Findings

Challenges around solitary eating

- **Few interruptions to disrupt escalating binge eating urges**
  Social interactions or unexpected events that could interrupt urges or restore awareness of eating were limited.
- **Fear of losing control over eating**
  Fear could begin before meals, including during food preparation. Repeated loss-of-control experiences could make eating alone distressing.
- **No one watching: an opportunity to binge**
  Some viewed being unobserved and uninterrupted as an opportunity to act on binge eating urges.

Designing for social presence and accountability

- **Feeling observed by an agent**
  Seeing the agent could make them more inclined to follow its guidance and less willing to be seen binge eating.
- **Observing their own eating during interaction**
  Monitoring themselves could increase awareness of their eating behavior.
- **Building a bond through repeated interaction**
  Some expected an ongoing relationship with the agent to strengthen motivation and accountability.

Designing just-in-time conversational support

Preferences differed across people and eating situations

TENSION 1
Proactive Support vs. User Agency

Initiation. AI-triggered or user-scheduled contact

Persistence. More directive support as urges rise; re-contact after refusal

Control. Choose whether to engage; avoid interruptions during ordinary meals

Access. Low-friction access to support; devices near the fridge or other relevant locations

TENSION 2
Eating-Focused Conversation vs. Attention Redirection

Direct engagement.
Before eating: meal planning and pre-commitment.
During/After eating: feedback or reflection on thoughts and emotions.

Attention redirection.
Talk about other topics when food-focused conversation could intensify unwanted thoughts or guilt around eating.

05 Design Implications — Design support that “adapts” within an eating episode

Coordinate when to initiate, what to discuss, and whether to continue as support needs and receptivity change.